

Michael Cardei

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Education

University of Virginia | Ph.D. in Computer Science August 2024 – Expected May 2028
Focus: Constrained Generative AI, Agentic AI, Diffusion Language Models, and AI for Science
NSF GRFP Fellow | **Advisor:** Professor Ferdinando Fioretto

University of Florida | B.S. in Computer Science | **GPA:** 3.92/4.00 June 2020 – May 2024

Research Experience

Graduate Research Assistant August 2024 – Present
University of Virginia, RAISE Lab, Advised by Professor Ferdinando Fioretto

- Conducting research on generative AI for scientific, engineering, and agentic applications, with an emphasis on responsible and constrained generation.
- Developing inference-time constrained generation methods for **diffusion language models**, integrating constrained optimization into the generative process to improve feasibility and controllability.

Graduate Student AI Research Intern May 2026 – August 2026
Lawrence Livermore National Lab, Livermore, California (On-site)

- Research **reinforcement learning-based** post-training methods to enhance large language models for **scientific discovery** and domain-specific reasoning.
- Developed and evaluated LLM post-training pipelines for scientific reasoning, analyzing model performance across domain-specific tasks.

Staff Machine Learning Research Scientist Intern May 2025 – August 2025
Visa, Foster City, California (On-site)

- Developed **Constrained Tabular Diffusion (CTDF)**, a training-free method for enforcing hard constraints during mixed-type tabular diffusion sampling in financial applications.
- Demonstrated zero constraint violations on large-scale financial datasets while preserving high data fidelity and improving utility in scarce-data settings.

Research Intern August 2023 – May 2024
University of Florida, Adaptive Learning and Optimization Lab, Advised by Dr. My Thai

- Investigated **privacy vulnerabilities** and explored implementation strategies within Federated Learning for **Large Language Models**.
- Examining neuron-based explainable AI methods for network intrusion anomaly detection mechanism analysis.

AI Research Intern (Robotics Institute Summer Scholars (RISS)) June 2023 – August 2023
Carnegie Mellon University Robotics Institute, ILIM Lab, Advised by Dr. Srinivasa Narasimhan

- Researched methods for context-driven road work-zone detection and localization for autonomous vehicles.
- Leveraged advanced **Computer Vision, Deep Learning, and NLP** techniques, including detection, instance segmentation, scene text recognition, and transfer learning.

Research Intern (REU) May 2022 – June 2023
Wake Forest University, Advised by Dr. Umit Topaloglu

- Researched **fairness and bias mitigation** in medical deep learning applications.
- Developed **privacy-preserving and federated learning** methods and evaluated robustness using adversarial attacks.

Publications

Conference Papers

- [Constrained Code Generation with Discrete Diffusion](#) 
M. Cardei*, L. Shao*, Z. Xie, W. Wang, and F. Fioretto.
Advances in Neural Information Processing Systems (NeurIPS) 2026.
- [Search-Augmented Masked Diffusion Models for Constrained Generation](#) 
M. Cardei*, H. Binh-Ta*, A. Velasquez, and F. Fioretto.
Advances in Neural Information Processing Systems (NeurIPS) 2026.
- [Constrained Discrete Diffusion](#) 
M. Cardei*, J. K. Christopher*, T. Hartvigsen, B. Kailkhura, and F. Fioretto.
Advances in Neural Information Processing Systems (NeurIPS) 2025.
- [Constrained Tabular Diffusion for Finance](#) 
M. Cardei, J. M. Munoz, O. Barrera, S. K. Chandrabhas, P. Saha.
ACM International Conference on AI in Finance (ICAIF) 2025
- [Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation](#) 
J. Christopher, **M. Cardei**, J. Liang, F. Fioretto.
2nd International Conference on Neuro-Symbolic Systems (NeuS) 2025
DARPA Disruptive Idea Award (Best Paper).
- [Speculative Diffusion Decoding: Accelerating Language Generation through Diffusion](#) 
J. Christopher, B. Bartoldson, T. Ben-Nun, **M. Cardei**, B. Kailkhura, F. Fioretto.
North American Chapter of the Association for Computational Linguistics (NAACL) 2025 **Oral.**
- [ROADWork Dataset: Learning to Recognize, Observe, Analyze and Drive Through Work Zones](#) 
A. Ghosh, R. Tamburo, S. Zheng, J. Alvarez-Padilla, H. Zhu, **M. Cardei**, N. Dunn, C. Mertz, S. Narasimhan.
International Conference on Computer Vision (ICCV) 2025.
- [Improving Equity in Deep Learning Medical Applications with the Gerchberg-Saxton Algorithm](#) 
S. Ay, **M. Cardei**, A. Meyer, W. Zhang and U. Topaloglu. *IEEE 11th International Conference on Healthcare Informatics (ICHI) 2023.*
- [Automatic Detection and Localization of Roadwork](#) 
S. Narasimhan, R. Tamburo, C. Mertz, D. Reddy, K. Vuong, A. Ghosh, S. Srivastava, N. Boloor, T. Ma, **M. Cardei**, N. Dunn, H. Zhu, Automatic Detection and Localization of Roadwork
Mobility21, Carnegie Mellon University, 2023.

Journal Articles

- [Improving Equity in Deep Learning Medical Applications with the Gerchberg-Saxton Algorithm](#) 
S. Ay, **M. Cardei**, A. Meyer, W. Zhang, and U. Topaloglu.
Journal of Healthcare Informatics Research (2024).

Workshop Papers

- [Constrained Molecular Generation with Discrete Diffusion for Drug Discovery](#) 
M. Cardei, J. K. Christopher, T. Hartvigsen, B. Kailkhura, and F. Fioretto.
AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development Workshop (NeurIPS 2025).
Best Student Paper Award.

Preprints / Under Review

- [Simple Self-Conditioning Adaptation for Masked Diffusion Models](#) 
M. Cardei*, H. Binh-Ta*, and F. Fioretto.

Selected Achievements and Awards

- National Science Foundation Graduate Research Fellowship Program (NSF GRFP), 2025
- NeurIPS Scholar Award, 2025
- Best Student Paper Award, NeurIPS AI4D3 Workshop, 2025, for “Constrained Molecular Generation with Discrete Diffusion for Drug Discovery”
- DARPA Disruptive Idea Award / Best Paper Award, NeuS, 2025, for “Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation”
- Best Paper Award, UVA LLM Workshop, 2024, for “Speculative Diffusion Decoding”

Teaching Assistant

CS 6501 Constrained-Aware Gen AI University of Virginia Spring 2026

CS4710 Artificial Intelligence University of Virginia Fall 2025

Academic Service

Reviewer

Advances in Neural Information Processing Systems (NeurIPS), 2026

ACM International Conference on AI in Finance (ICAIF), 2025

Relevant Courses

University of Virginia (Ph.D.)

Graph Machine Learning, Natural Language Processing, Machine Learning in Systems Security, Machine Learning for Image Analysis, Machine Learning for Software Reliability

University of Florida (B.S.)

Trustworthy Machine Learning (Graduate Course), Applied Machine Learning, Natural Language Processing, Introduction to Multi-Modal Machine Learning, Programming Language Concepts, Engineering Statistics, Operating Systems, Data Structures and Algorithms

Skills

- **Research:** Generative AI, Agentic AI, Large Language Models, LLM Post-Training, Diffusion Language Models, Diffusion Models, Reinforcement Learning for LLMs, Deep Learning, Machine Learning, Computer Vision, Federated Learning, Constrained Generation, Applied AI
- **Languages:** Python, C++, Java, SQL
- **Tools/Frameworks:** PyTorch, TensorFlow, Keras, vLLM, LoRA, Scikit-Learn, TensorFlow Federated, GitHub, Hugging Face Transformers, Weights & Biases